out only a little bit of total entropy from the system. Many cosmologists take the opposite approach and say that it's easier to fluctuate to a still relatively high-entropy state than to create a pocket where the entropy is very, very low. Settling this question could give us a handle on one scenario for the origin of the entire cosmos, as well as lend us some peace of mind with respect to our possible fate of infinitely playing back our most cringe-worthy moments forever.

And for some cosmologists, understanding how we started with a low-entropy state in the early universe, and determining once and for all whether or not we have to worry about Boltzmann Brains or Poincaré recurrences, are questions that shake the very foundations of our cosmological model. Trying to find a way to set up a low-entropy initial state has prompted some to hypothesize entirely new cosmic histories (as we'll discuss in Chapter 7), though the issue is very far from resolved. And the possibility of fluctuations is so disturbing to our picture of a sensible cosmos that it has been described by Sean Carroll as "cognitively unstable." It's not that it can't be true, but that if it is, nothing makes sense, and we might as well give up on trying to understand the universe at all. The jury is still out on this one.

If you're not too disturbed by the possibility of disembodied sentient brains popping into and out of existence, the possibility of rare random fluctuations can, in some sense, drag some order out of the Heat Death's nihilistic disarray. But even in this most optimistic view, a universe dominated by a cosmological constant unquestionably spells doom for any beings living within it, as absolutely every coherent structure is destined for dark, lonely emptiness and decay. Before dark energy was discovered, physicists like Freeman Dyson came up with speculative proposals that a machine whose computation constantly slows can persist for an arbitrarily long time into the cosmic future.[fn12] But even this ideal machine would be subject to entropic erosion via the Second Law, and would eventually disintegrate into waste heat in the face of the de Sitter horizon. The timescales for the achievement of maximum entropy—the true and timeless Heat Death—depend on estimates of the decay time of the proton, which are still uncertain. Nonetheless, we probably have a good 10^{1000} years or so before we and all other thinking structures fade from the possibility of memory.

It could be worse.